Strain-Tensor Eigenvalue Decomposition and Intermediate Alignment in 3D Navier–Stokes

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Abstract

We record the elementary algebraic decomposition of the symmetric strain-rate tensor of a divergence-free velocity field and its consequences for the local vortex-stretching term. Because the trace of the strain vanishes, the intermediate eigenvalue controls the leading contribution once the vorticity direction is aligned with the intermediate eigenvector. The identities are pointwise and unconditional, and the trace identity and the intermediate-eigenvalue representation are checked in Lean 4 (Section 5); the geometric and analytic discussion is not formalized. The identities do not by themselves produce an a priori bound on the L^∞ norm of the strain and therefore leave the supercritical gap (Hypothesis B) open.

1 Eigenvalue structure of the strain tensor

Let u be a smooth divergence-free vector field on $\mathbb{T}^3$ and let

$$S = \frac{1}{2}\big(\nabla u + (\nabla u)^T\big)$$

be the symmetric strain-rate tensor. Then $\operatorname{tr} S = \nabla \cdot u = 0$, so the three real eigenvalues satisfy

$$\lambda_1 \le \lambda_2 \le \lambda_3, \qquad \lambda_1 + \lambda_2 + \lambda_3 = 0. \tag{1}$$

At each point where S is continuous there exists an orthonormal eigenbasis $\{e_1, e_2, e_3\}$. Writing the unit vorticity direction

$$\boldsymbol{\xi} = \frac{\omega}{|\omega|} = \xi_1 e_1 + \xi_2 e_2 + \xi_3 e_3$$

(with $\xi_1^2 + \xi_2^2 + \xi_3^2 = 1$) one obtains the quadratic form

$$\boldsymbol{\xi} \cdot S\boldsymbol{\xi} = \lambda_1 \xi_1^2 + \lambda_2 \xi_2^2 + \lambda_3 \xi_3^2. \tag{2}$$

Proposition 1.1 (Intermediate-eigenvalue representation)**.**

$$\boldsymbol{\xi} \cdot S\boldsymbol{\xi} = \lambda_2 + (\lambda_1 - \lambda_2)\xi_1^2 + (\lambda_3 - \lambda_2)\xi_3^2.$$

Proof. Substitute $\xi_2^2 = 1 - \xi_1^2 - \xi_3^2$ into (2) and rearrange. □

2 Geometric interpretation

Because $\lambda_1 \le 0 \le \lambda_3$, the deviation terms $(\lambda_1 - \lambda_2)\xi_1^2$ and $(\lambda_3 - \lambda_2)\xi_3^2$ measure the departure of $\boldsymbol{\xi}$ from the intermediate axis e_2. Classical geometric criteria (Neustupa–Penel, Miller) observe that intense stretching in turbulent flows correlates with preferential alignment of $\boldsymbol{\xi}$ with e_2 rather than with the most extensional eigenvector e_3. The algebraic identity above makes this statement precise: the leading contribution is exactly λ_2, while the remainder is controlled by the angular dispersion of $\boldsymbol{\xi}$.

3 Relation to the alignment kernel

Under a Hölder condition

$$|\boldsymbol{\xi}(x) - \boldsymbol{\xi}(y)| \le K|x-y|^{\alpha}, \qquad \alpha \ge \frac{1}{2},$$

on high-vorticity sets, the Constantin–Fefferman singular integral that represents $\boldsymbol{\xi} \cdot S\boldsymbol{\xi}$ desingularizes. Heuristically, the eigenvalue decomposition isolates the same phenomenon; we do *not* prove here that the integral of the deviation terms is controlled by the angular dispersion functional

$$\mathcal{D}_{\boldsymbol{\xi}}(t) = \iint_{\Omega_M \times \Omega_M} |\omega(x)|^2 |\omega(y)|^2 |\boldsymbol{\xi}(x) \times \boldsymbol{\xi}(y)|^2 \, K_{\mathrm{BS}}(x-y)\, dx\, dy,$$

and this comparison is stated only as motivation.

4 Conditional status

The identities of this note are pointwise and algebraic. They do not imply integrability of $\|S\|_{L^\infty}$ in time. In particular, they do not prove Hypothesis B

$$\int_0^T \|S(t)\|_{L^\infty}\, dt < \infty$$

for arbitrary large smooth data. The supercritical gap between the energy inequality and the Beale–Kato–Majda criterion remains open.

5 Lean formalization: exact scope

The repository `https://github.com/BenFrohman/frohmanian-strain-eigenvalue` contains `StrainEigenvalue.lean` (Lean 4.30.0-rc1, Mathlib `v4.30.0-rc1`, commit `0c154d6`). A `StrainTensor` is a real symmetric 3×3 matrix with zero trace. Lean indices $0, 1, 2$ play the roles of $1, 2, 3$ here. Mathlib's eigenvalue indexing is *not* proved to be sorted, so the ordering $\lambda_1 \le \lambda_2 \le \lambda_3$ and the signs $\lambda_1 \le 0 \le \lambda_3$ are not used or proved in Lean. The kernel checks the following, using only the axioms `propext`, `Classical.choice`, `Quot.sound`:

- `eigenvalues_sum_zero`: $\lambda_1 + \lambda_2 + \lambda_3 = 0$.
- `quadratic_form_eq_sum`: with U the orthonormal eigenvector matrix from Mathlib's spectral theorem and $\eta = U^T\xi$, $\xi \cdot S\xi = \sum_i \lambda_i \eta_i^2$ for every $\xi \in \mathbb{R}^3$ (equation (2)).
- `sum_sq_eigenCoords`: $\sum_i \eta_i^2 = \sum_i \xi_i^2$.
- `quadratic_form_intermediate`: if $\sum_i \xi_i^2 = 1$, then $\xi \cdot S\xi = \lambda_2 + (\lambda_1 - \lambda_2)\eta_1^2 + (\lambda_3 - \lambda_2)\eta_3^2$. This is Proposition 1.1.
- Non-vacuity: $S = \mathrm{diag}(1, -1, 0)$ is a `StrainTensor` with eigenvalues not all zero, and for $\xi = (1, 0, 0)$ both sides of Proposition 1.1 equal 1.

The theorem `intermediate_representation` proves the same identity for `stretching`, which is $\sum_i \lambda_i \xi_i^2$ with ξ already in eigen-coordinates. An earlier skeleton gave `stretching` the hypothesis $\|\xi\| = 1$. On $\mathrm{Fin}\, 3 \to \mathbb{R}$, Mathlib's $\|\cdot\|$ is the sup norm, so that does not follow from $\sum_i \xi_i^2 = 1$: $\xi = (3/5, 4/5, 0)$ has $\sum \xi_i^2 = 1$ but $\|\xi\|_\infty = 4/5$. Lean checks this counterexample (`sum_sq_eq_one_not_imp_norm_eq_one`). The hypothesis is now $\sum_i \xi_i^2 = 1$. The development has no `sorry`.

Nothing about vorticity, Navier–Stokes dynamics, Hölder alignment, the Constantin–Fefferman kernel, $\mathcal{D}_{\boldsymbol{\xi}}$, or Hypothesis B is formalized. A kernel check establishes the algebraic statements above and nothing more.

References

[1] P. Constantin and C. Fefferman, Direction of vorticity and the problem of global regularity for the Navier–Stokes equations, Indiana Univ. Math. J. **42** (1993), 775–789.

[2] J. Neustupa and P. Penel, Regularity of a suitable weak solution to the Navier–Stokes equations as a consequence of the regularity of one velocity component, in *Applied Nonlinear Analysis*, Kluwer, 1999, pp. 391–402.